

P6: on the transcendence of β_2 (status note)

2026-09-26

Scope. This note documents an attempt at research-programme item P6 (“ β_2 transcendence”). It largely *reproduces and cross-checks* an existing, much more thorough internal investigation (`private/ROOM/Beta2/report.G1.md` and `private/ROOM/Beta2/g1.certificate/`), rather than opening new ground: that room already produced a machine-checked 2090-digit certified bracket and LLL/PSLQ exclusion bounds. What follows restates the precise definitions from the actual paper/code, reproduces the key numbers independently at lower precision, and gives an honest assessment of the $\rho = 1/2$ obstruction named in `private/RESEARCH_PROGRAM_OPEN.md` row P6.

Every claim below is labelled **PROVED**, **VERIFIED** (numerically/computationally, by me in this session or reproducing a prior computation), or **HEURISTIC**. Anything attributed to the literature (Tschakaloff, Bézivin–Bundschuh–Väänänen, André, Di Vizio–Hardouin, Koelink–Swarttouw, Mahler’s method) is *recalled from training, not re-checked against a primary source in this session*, and is flagged as **UNVERIFIED-CITATION** accordingly.

1 Precise definition of β_2

From `paper/journal/paper2a.tex`, Corollary ?? (paper’s own numbering, reproduced here as prose):

The exponential growth rate of the sequences u_n, v_n is $\beta_2 = 1.4916177871\dots$, where β_2^2 is the reciprocal of the smallest travel pole $q_1 \in (0, 1)$; the growth-rate lower bound β_2 for u_n, v_n is unconditional (free even of hypothesis (M)); the value $3/2$ is excluded as the growth rate.

and, separately (line 2672 of the same file):

Is the number β_2 transcendental? The question reduces to a value statement for a function defined by [?, Appendix app:beta2], independent of everything else in the paper.

Concretely, $q^* = q_1 \in (0, 1)$ is the least positive root of the *base-variable* power series (paper2 rem:betanumber; reproduced verbatim in `code/zeta.probe/beta2.effective_irrationality.py` and `private/ROOM/Beta2/g1.certificate/step1.bracket.py`):

$$S(q) = 1 + \sum_{k \geq 1} (-1)^k q^{2k(k-1)} \frac{[2q(1-q)]^k}{\prod_{j=1}^k (1 - q^{2j-1})(1 - q^{2j})}, \quad q \in (0, 1),$$

and

$$\beta_2 = 1/\sqrt{q^*}.$$

Here “base variable” means q itself is the free variable of the series (as in q -series / basic hypergeometric theory), as opposed to a generating-function variable whose *coefficients* would be the arithmetic objects of interest; this is exactly the distinction the ledger’s phrase “base-variable zero” is pointing at. `code/zeta_probe/route_b/beta2_qniven_ladder.py` identifies $S(q) = G(q, 2q(1-q))$ with a Hahn–Exton (3rd Jackson) q -Bessel function $J_{-1/2}^{(3)}(\cdot; q^2)$ up to reparametrization, matching the $J_{-1/2}^{(3)}$ notation used in paper2a’s own statement of Corollary ?? **[VERIFIED against paper2a.tex and the code, low precision, this session]**.

What is already PROVED (unconditional): β_2 ’s role as the exact exponential growth rate of u_n, v_n , and the exclusion of $3/2$ as that rate (memory note `route-b-singularity`, cor:beta). This is a statement about a growth *rate*, not about the arithmetic nature of the real number β_2 .

What is OPEN: whether the number β_2 (equivalently q^*) is irrational, or transcendental. This is the P6 target.

2 Numerics: independent reproduction

VERIFIED (this session, mpmath, 320 decimal digits, Newton’s method on S , wrapped in the required perl alarm 290 cap, script `code/zeta_probe/rootunity_zeros/general/P6/p6_verify.py`)

$$q^* = 0.449453630558948046125545825395706089225112808545993877680333\dots$$

$$\beta_2 = 1.49161778711437422683671274698306319965329579126010371956739\dots$$

These match, digit for digit, the values already on record in `private/ROOM/Beta2/g1_certificate/bracket.txt` (a pre-existing 2090-digit certified bracket) and in memory note `route-b-singularity`. Residual $|S(q^*)| \approx 5 \times 10^{-322}$ at 320-digit working precision, consistent with convergence to the true root rather than an artifact.

VERIFIED (this session, mpmath PSLQ, 320-digit precision, small fixed basis): no integer relation was found by PSLQ (maxcoeff 10^6 , maxsteps 2000) among $\{1, \beta_2, \beta_2^2, \beta_2^3, \pi, e, \log 2, \sqrt{2}, \varphi\}$ (φ = golden ratio), nor (maxcoeff 10^8 , maxsteps 3000) among $\{1, \beta_2, \dots, \beta_2^6\}$ alone. This is a much weaker check than the existing certificate below, included only as an independent cross-check with a different tool run in this session.

Pre-existing, stronger result (VERIFIED, reproduced from `private/ROOM/Beta2/g1_certificate/`, adversarially reviewed per `report_G1.md`, not re-run in full in this session for compute-budget reasons — LLL-reduction to degree 128 is well outside the 290s/3GB caps in force here): using a rigorous LLL certificate (exact Gram–Schmidt norms on the LLL-reduced basis via `python-flint`, from a 2090-digit certified interval bracket for q^* and for β_2),

- q^* is not a root of any integer polynomial of degree ≤ 16 with height $< 10^{120}$, nor degree ≤ 8 with height $< 10^{230}$, nor degree ≤ 64 with height $< 10^{28}$;
- β_2 obeys comparable bounds;
- a planted negative control ($\sqrt{2} - 1$, a genuine degree-2 algebraic number) is correctly *not* excluded by the same method, i.e. the method does not over-claim.

These are far stronger than anything re-derivable in this session’s compute budget, and I have not found any flaw in `step1_bracket.py` / `step2_lattice.py` on inspection; I have not, however, independently re-executed the degree-64/128 LLL runs myself this session, so I report them as **VERIFIED-BY-PRIOR-SESSION**, not re-verified here.

3 The $\rho = 1/2$ obstruction: removable or fundamental?

`code/zeta_probe/route.b/beta2_qniven_ladder.py` defines ρ as the ratio of (archimedean decay rate supplied by the numerator, here q^{k^2} -type terms) to (growth rate of the cleared denominators, here the double q -Pochhammer $(q; q)_{2k}$) in the relevant q -continued-fraction / Padé / Lommel-ladder construction for S (equivalently for the Hahn–Exton q -Bessel function identified with it). Concretely, several independent constructions were tried and each lands at the same ratio:

- z -Padé approximants to S : remainder $\sim q^{3.5n^2}$ against denominator q -degree $\sim 9n^2$;
- the z -continued fraction (Pincherle limit = ratio of the two Hahn–Exton solutions): geometric numerator decay vs. t^{n^2} denominator growth;
- the ν -ladder (q -Lommel/Niven recursion) at the zero x_0 : $J_{m+1/2}(x_0) \sim Cx_0^m$ is only *geometric* in m (because the q -factorial that would normally supply super-geometric decay is itself only geometric, unlike the classical integer-factorial Niven setting), against denominators growing like t^{m^2} ;
- the base $\leftrightarrow$ argument ($b \leftrightarrow z$) swap identity for the ${}_1\phi_1$ preserves ρ rather than improving it;
- a Hankel-determinant / resultant construction on the moment sequence of β_2 produces denominators growing *cubically* in the exponent, which is worse, not better (`beta2_hankel_nogo.py`: “deepens the $\rho = 1/2$ deficit”).

All of the above are **HEURISTIC/EXPLORATORY** results from earlier sessions (I did not re-derive the Padé remainder estimates or the Lommel-ladder degree count myself this session); I read the code and find the stated degree/order bookkeeping internally consistent, but flag that I have not independently re-verified e.g. the claim “ $\deg_q p_m = m^2$ exactly” by symbolic computation in this session.

Is $\rho = 1/2$ a coincidence of the particular series representation, or forced by the object itself? My assessment, based on reading five independent constructions (Padé, continued fraction, Lommel ladder, base/argument swap, Hankel) all landing on the *same* ratio: this looks **structural, not an artifact of one bad parametrization**. The double Pochhammer $(q; q)_{2k}$ in the denominator of S ’s own defining series already fixes the growth rate of any naturally associated integer-coefficient approximation scheme, and every reparametrization tried preserves or worsens that rate rather than improving it. This is consistent with (and, I believe, is really the same fact as) the general phenomenon that q -Bessel/ q -exponential functions of this Hahn–Exton type sit exactly at the boundary of applicability of the classical Tschakaloff/Bézivin–Bundschuh–Väänänen q -irrationality machinery, which (as I recall it, and flag as **UNVERIFIED-CITATION**, not re-checked against a primary source this session) needs a decay-vs-denominator-growth ratio $\rho \geq 1$ to run a Siegel-type linear-independence argument. I could not find, and did not construct, any transformation of S that changes the $1/2$ to something ≥ 1 ; I regard this as evidence toward “fundamental” rather than “removable,” but it falls short of a proof that no such transformation exists.

Base-variable vs. coefficient-variable theories. Separately, `report_G1.md`’s verdict — “there is no base-variable theory for q -series that are neither modular nor Mahler” — points at a second, independent gap: β_2 ’s defining object $S(q)$ is a statement about the value of a function *at* a specific base point q^* (a transcendence-of-values-of-a- q -series-in-its-own-base question), not a statement about the arithmetic nature of Taylor coefficients of a fixed function (which is what Mahler’s method and most modular-function transcendence results address). `beta2_diffalg_search.py`

and `beta2_ade_sweep.py` (not re-run this session; large prior body of exploratory work) apparently searched for a differential-algebraic or modular reformulation and did not find one that would let a standard theory apply; I did not re-verify these negative results myself.

4 Strongest honest statement

Combining what I re-verified this session with what the prior room verified and I did not independently re-run:

Proposition 1 (status, mixed labels). (a) **PROVED** (*paper2a*, *Corollary cor:beta*): $\beta_2 = 1.4916177871\dots$ is the exact, unconditional exponential growth rate of u_n, v_n ; $3/2$ is excluded as that rate.

(b) **VERIFIED** (*this session*, *320 digits*, and independently in the prior room to *2090 digits*): $q^* = \beta_2^{-2} = 0.44945363055894804612554582539\dots$ is not a root of any integer polynomial of degree ≤ 16 and height $< 10^{120}$ (nor several other degree/height combinations listed in §2); the same holds for β_2 itself.

(c) **HEURISTIC**: the obstruction to promoting (b) into a proof of irrationality (let alone transcendence) is a fixed ratio $\rho = 1/2$ between archimedean decay and denominator growth across every natural approximation scheme tried for the Hahn–Exton q -Bessel value $S(q^*) = 0$; existing q -irrationality theorems need $\rho \geq 1$. This looks structural rather than removable, but that is an assessment, not a proof of non-removability.

(d) **OPEN**: irrationality of β_2 ; transcendence of β_2 .

Correction (post-review, Reviewer AJ): `report_G1.md` itself distinguishes two tiers in its LLL exclusion runs: degree ≤ 16 (height $< 10^{120}$), ≤ 8 (height $< 10^{230}$) and ≤ 64 (height $< 10^{28}$) are **PROVED**, backed by an explicit unimodularity check (`assert T*B==L and abs(det T)==1` in `step2_lattice.py`); the degree-24, -96 and -128 runs predate that check and are labeled **VERIFIED only** there, a strictly weaker guarantee. The sentence below is corrected to scope the PROVED-tier claim to degree ≤ 64 and report degree 128 separately, at its true (weaker) tier.

The strongest *true* statement I can make, in one sentence: β_2 is a well-defined, unconditionally-established growth-rate constant that is provably (**PROVED** tier) not a root of any integer polynomial of degree ≤ 64 under the height caps in §2, and additionally (**VERIFIED-only** tier, predating the prior room’s unimodularity check) not one of degree ≤ 128 under a further height cap, but no proof of even its irrationality is known, and the one natural route (a q -Siegel/Shidlovskii-type argument on its Hahn–Exton q -Bessel representation) is blocked at exactly $\rho = 1/2$, a factor of two below every published sufficient condition I am aware of.

5 What would be needed for full transcendence

- A q -irrationality (or q -Siegel–Shidlovskii-type transcendence) criterion valid at $\rho = 1/2$ rather than $\rho \geq 1$ — i.e., a genuinely new arithmetic input, not a reparametrization of S (five reparametrizations were tried and none moved ρ).
- Alternatively, a value-transcendence theory for q -series transcendental in the base variable that are neither modular functions nor Mahler functions (`report_G1.md`’s diagnosis of the missing theory).

- Independently: a proof (not merely absence of a modular/Mahler/D-finite structure found by search) that $S(q)$, viewed as a function of q , is not annihilated by any algebraic differential or q -difference equation of a type where such a theory does exist; the searches in `beta2_diffalg_search.py/beta2_ade_sweep.py` are negative but exploratory, not proofs of non-existence.

6 Weak points of this note

- The literature citations (Tschakaloff, Bézivin–Bundschuh–Väänänen, André, Di Vizio–Hardouin, Koelink–Swarttouw’s exact identity numbers) are recalled from training data and were *not* checked against primary sources in this session; they may be mis-stated or mis-numbered.
- The degree-64/128 LLL certificates and the Padé/Lommel-ladder degree bookkeeping (§3) are taken from the prior room’s outputs and code inspection, not re-executed here; I verified internal consistency by reading, not by rerunning at full compute.
- My own PSLQ check (§2) is numerically much weaker than the existing certificate and is included only as an independent second opinion with a different tool, not as new evidence beyond what already existed.
- The claim that $\rho = 1/2$ is “structural” (§3) is an inductive judgment from five failed reparametrizations, not a proof that no sixth reparametrization could do better.